GWCD Documentation - The Public Application

Vermont AHS GIS and Data Viz | Groundwater Chemistry Database (GWCD)

The Public Application

Drinking Water Quality — ArcGIS Experience Builder app, owned by `VDH_EnvHealth`, public.

This was verified directly against the app's live Experience Builder configuration (not just inferred from the ETL script), by pulling the item's JSON config from the ArcGIS REST API.

Data source

The app's map/table/search widgets are built on:

```
GroundwaterChemistry_GWCDPointTableExternal  (joined to)  GroundwaterChemistry_TownBoundaries
```

This confirms `GWCDPointTableExternal` is in fact what's wired to the public product — i.e., the suppressed/deduplicated table described in 05-geocoding.md, not the unsuppressed `GWCDPoints` table.

Fields used in the app

Matches the ETL script's output fields exactly:

- `AnalyteAlias` (displayed as "Analyte")
- `TOWNNAME` / `TOWNNAMEMC` (displayed as "Town")
- `CityCount` (displayed as "Total Tests")
- `CityAboveMCL`, `CityAboveSMCL`, `CityAboveVHA`, `CityAboveThreshold`
- `PctAboveMCL`, `PctAboveSMCL`, `PctAboveVHA`, `PctAboveThreshold`
- `MapMean`, `MapMedian`, `MapMin`, `MapMax`, `MapPercentile95`, `MapStdDev`, `MapMDL`, `MapPopupHeading` (pre-formatted strings for map popups)

Analytes currently surfaced

The app only has a dedicated map + summary table + search widget set for six analytes — exactly the six the ETL script generates popup text for:

1. Arsenic
2. Fluoride

3. Gross Alpha
4. Nitrate
5. Radon
6. Uranium

All other analytes in the `exceedanceDictionary` (Lead, Manganese, Copper, Iron, Sodium, Chloride, Radium, TDS, Turbidity, Odor, E.coli, Total Coliform, pH, Hardness, Nitrite, Total Nitrate and Nitrite) **are calculated** in the pipeline (exceedance flags, town-level stats all exist in `GroundwaterChemistry_GWCDPoints`) but are **not currently displayed** in this app. A grant-reporting dashboard could surface any of these directly from `GWCDPoints` without needing new calculation logic.

Layout

Each analyte has its own sidebar (full-screen) with: a text panel citing its MCL/SMCL/VHA value, a map, a controller toggling between map and table view, a summary table, and an address search widget.